Lecture 4:
anatomy of Bayes' theorem

September 2, 2026

let's take apart Bayes' theorem

$$p(\theta \mid X) = \frac{p(\theta)\, p(X \mid \theta)}{p(X)}$$

θ: parameters of your model

X: data

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$p(\theta \mid X)$: **posterior** — updated belief about θ

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probability distribution over X

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likelihood from the jupyter notebook photon count demo

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if you change your noise model, you change the likelihood

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Gaussian likelihood often due to CLT, but not always! (Poisson, binomial, etc)

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what you believe about θ before this data: *e.g.* a previous measurement, theoretical constraints, physical bounds (*e.g.* $M_\nu > 0$)

not a fudge factor! as principled a modeling choice as the likelihood and requires justification

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not a fudge factor! as principled a modeling choice as the likelihood and requires justification **also unavoidable**

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a prior flat in θ is **not** flat in θ^2 or $\log \theta$

two people can fit the same model with different parameterizations, both using a "flat, uninformative" prior and get different posteriors

use flat priors with caution, also here are some alternatives…

the **Jeffreys prior** — general purpose non-informative priors

Harold Jeffreys — geophysicists who hated continental drift and plate tectonics, oops

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Fisher information

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Fisher information

$$= \int dX \; p(X|\theta) \left(\frac{\partial \log p(X|\theta)}{\partial \theta}\right)^2$$

$\approx$*"how strongly does the log-likelihood depend on θ"*

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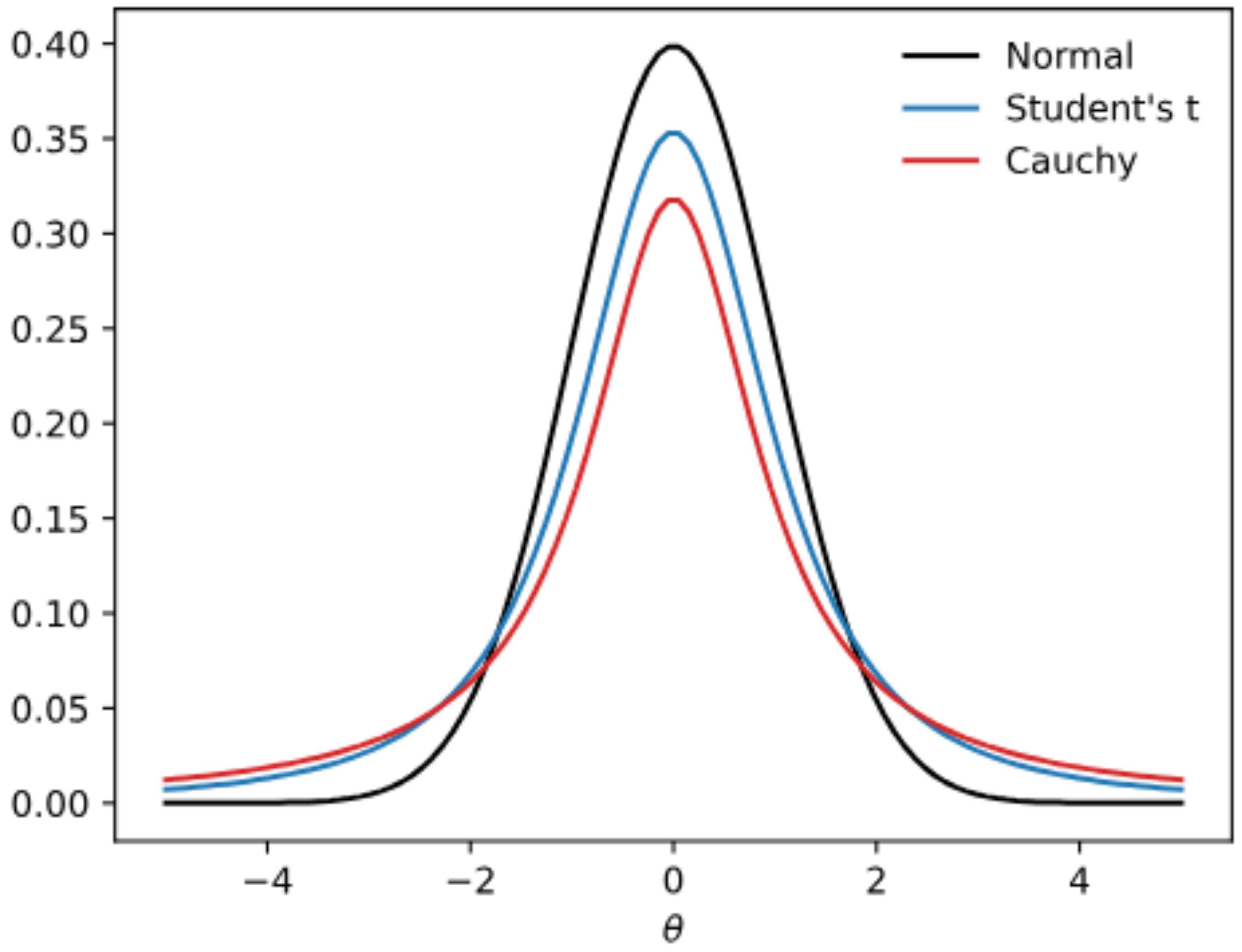
robust priors — not very confident in our priors so we want to reduce its influence on the results



prior distributions with heavy tails (student-t, cauchy) allow for data that are far from the prior mean

image credit: P. Melchior

jupyter notebook demo — *https://github.com/changhoonhahn/ast384c/blob/main/lectures/Lec4_bayes_demo.ipynb*

in-class

1. Repeat "Example 1: Less Simple Photon Count" for three priors: flat prior, $\sigma = 12 \pm 2$, and an overconfident prior $\sigma = 12 \pm 0.5$

2. Reduce to $N = 5$ observations (from $N = 100$) and repeat for the three priors

3. $N = 1000$?

How do the posteriors in (2) and (3) compare to in (1)?

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*with enough data, the likelihood dominates — but with a handful of noisy points, you may be **prior dominated**.* when in doubt →**sensitivity analysis**

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often very difficult to compute: *curse of dimensionality, likelihood is concentrated in the prior volume, multi-modality* 🙃 **why do we care about it, ever?**

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choice of model $\mathscr{H}$

$$p(\theta \mid X, \boxed{\mathscr{H}}) = \frac{p(\theta \mid \mathscr{H})\, p(X \mid \theta, \mathscr{H})}{p(X \mid \mathscr{H})}$$

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$p(X \mid \mathscr{H})$ — *how well does $\mathscr{H}$ explain the data X, integrated over the prior*

model comparison: $p(X \mid \mathscr{H}_1)$ versus $p(X \mid \mathscr{H}_2)$ — Bayes factor

posterior predictive distribution — predicting the future with Bayesian reasoning

predict future observations based on the observe data

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$$p(x \mid X) = \int d\theta \; p(x \mid \theta) \; p(\theta \mid X) = \mathbb{E}_{p(\theta|X)} \left[p(X \mid \theta) \right]$$

assignment:

probabilistically forecast F_{new} for a follow up experiment to "Example 1: Less Simple Photon Count" — *i.e.* plot the posterior predictive distribution

posterior predictive: $$p(x\,|\,X) = \int d\theta\; p(x\,|\,\theta)\; p(\theta\,|\,X)$$

don't forget to normalize the posterior

next class: **optimization**

what actually happens inside `scipy.optimize`